

In both single and multiple-dose studies in healthy subjects, once daily 500 mg (2x500 mg tablets) dosing provided equivalent systemic exposure, as measured by area-under-the-curve (AUC), and up to 35% higher C_{max} of metformin relative to the immediate release given as 500 mg twice daily. Metformin hydrochloride extended-release tablets are not recommended immediately after meals to maximize therapeutic benefit.

Single oral doses of metformin hydrochloride extended-release tablets from 500 mg to 2500 mg resulted in less than proportional increases in both AUC and C_{max} . Use of the high-dose tablets increased the systemic exposure (as measured by AUC) from metformin hydrochloride extended-release tablets by about 30% and 73%, respectively, relative to fasting. Both men and postmenopausal women, by approximately 6 hours but not 12 hours, was not affected.

In a two-way, single-dose crossover study in healthy subjects, the 500 mg tablet was found to be bioequivalent to two 500 mg tablets under fed conditions based on equivalent C_{max} and AUCs for the two formulations.

Distribution

The mean total volume of distribution (V_d) of metformin following single oral doses of 500 mg immediate release metformin hydrochloride averaged 64 ± 35L. Metformin is negligibly bound to plasma proteins. Metformin partitions into erythrocytes, most likely as a function of time. At usual clinical doses and dosing schedules, metformin does not alter state plasma concentrations of erythrocytes are reached within 24 to 48 hours and are generally < 1 mg/mL. During controlled clinical trials, which served as the basis of approval for metformin, maximum metformin plasma levels did not exceed 5 mg/mL, even at maximum doses.

Metabolism

Interventions single-dose studies in healthy subjects demonstrate that metformin is excreted unchanged in the urine, not does not undergo hepatic metabolism (no metabolite lines have been identified in humans), nor urinary excretion. Metabolism studies with extended-release metformin but have not been conducted.

Excretion

Renal clearance is approximately 3.5 times greater than creatinine clearance, which indicates that tubular secretion is the major route of metformin elimination. Following oral administration, approximately 90% of the absorbed drug is eliminated via the renal route within the first 24 hours, with a plasma elimination half-life of approximately 6.2 hours. In black, the elimination half-life is approximately 17.6 hours, suggesting that the erythrocyte mass may be a component of distribution.

12.4 Specific Populations

Renal impairment

Following a single dose administration of metformin hydrochloride extended-release tablets, 500 mg in patients with mild and moderate renal failure (based on measured creatinine clearance), the oral and renal clearance of metformin were decreased by 52% and 50% and 15% and 53%, respectively (See **WARNINGS AND PRECAUTIONS (3)**). Metformin peak and systemic exposure was 27% and 6 % greater, respectively in mild renal impairment and 74% and 2.3% fold greater in moderate renal impairment patients as compared to healthy subjects.

Use of metformin in patients with renal impairment increases the risk for lactic acidosis.

Metformin hydrochloride extended-release tablets are contraindicated in patients with renal impairment. (See **CONTRAINDICATIONS (4)** and **WARNINGS AND PRECAUTIONS (3.2)**).

Hepatic impairment

No pharmacokinetic studies of metformin hydrochloride extended-release tablets have been conducted in subjects with hepatic impairment. Use of metformin hydrochloride tablets with hepatic impairment has been associated with some cases of lactic acidosis. Metformin hydrochloride extended-release tablets are not recommended in patients with hepatic impairment. (See **WARNINGS AND PRECAUTIONS (3.3)**).

Geriatrics

Limited data from controlled pharmacokinetic studies of metformin hydrochloride in healthy subjects suggest that total plasma clearance of metformin is decreased by 35%, the half-life is prolonged by 64%, and C_{max} is increased by 78%, compared to healthy young subjects. From these data, it appears that the change in metformin pharmacokinetics with aging is primarily accounted for by a change in renal function. Metformin treatment should not be initiated in patients of any age or use metformin of extended-release tablets from these data that the renal function is normal. (See **WARNINGS AND PRECAUTIONS (3.3)** and **DOSE AND ADMINISTRATION (2.1)**).

Gender

In the pharmacokinetic studies in healthy volunteers, there were no important differences between male and female subjects with respect to metformin AUC and C_{max} . However, C_{min} for men was 40% higher in female subjects as compared to males. The gender differences for C_{min} is unlikely to be clinically important. Similarly, in controlled clinical studies in patients with type 2 diabetes, the antihyperglycemic effect of metformin hydrochloride tablets was comparable in males and females.

Race

There were no definitive conclusions on the differences between the races with respect to the pharmacokinetics of metformin because of the imbalance in the respective sizes of the racial groups. However, the data suggest a trend toward higher metformin C_{min} or AUC values for metformin in African subjects relative to Caucasians, Hispanic and Black subjects. The differences between the Asian and Caucasian groups are unlikely to be clinically important. In controlled clinical studies of metformin hydrochloride in patients with type 2 diabetes, the antihyperglycemic effect was comparable in whites ($n = 249$), blacks ($n = 51$) and Hispanics ($n = 24$).

Pediatrics

No pharmacokinetic data from studies of metformin hydrochloride extended-release tablets in pediatric subjects are available.

12.5 Drug Interactions

Pharmacokinetic drug-drug interaction studies with metformin hydrochloride extended-release tablets have not been performed except for one with glyburide. However, such studies have been performed on metformin.

Table 2: Effect of Coadministered Drug on Plasma Metformin Systemic Exposure

Coadministered Drug ^a	Dose of Coadministered Drug ^a	Dose of Metformin ^b	Geometric Mean Ratio (ratio with/without coadministered drug) No effect = 1.00
			AUC ^c
No dosing adjustments required for the following:			
Glyburide	5 mg	500 mg ^d	0.99 ^e
Chlorzoxazone	400 mg	850 mg	1.09 ^e
Nifedipine	10 mg	850 mg	1.16 ^e
Propranolol	40 mg	850 mg	0.90 ^e
Warfarin	400 mg	850 mg	1.02 ^e

^a Coadministered drug was given as a single dose.

^b Metformin hydrochloride extended-release tablets were given as a single dose.

^c AUC = AUC_{0-24h}.

^d Metformin hydrochloride extended-release tablets were given as a single dose.

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01/11/2012

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